

Bayesian Compositional Time-Series Modeling for Predicting Daily Avoidable Deaths for Emergency Departments in Bristol, England

SPHERE-PPL Forecasting Contest — NHS Acute Patient Harm
Forecasting Submission

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June 5, 2026

Model Specification

Overview

We developed a Bayesian compositional time-series model to forecast daily estimated avoidable deaths associated with emergency department admission delays in the Bristol healthcare system. It decomposes the $\log_{1p}$ -transformed avoidable deaths into four distinct components: a long-term cyclical seasonal baseline, a regularized distributed-lag linear covariate effect, a day-of-week effect, and a latent autoregressive process representing short-term system memory. Parameters across all components are estimated jointly, and their relative strength can be regularised in the Bayesian framework. The model is implemented in PyMC and its posterior distribution is sampled using the No-U-Turn Sampler (NUTS) [Hoffman and Gelman, 2014, Abril-Pla et al., 2023].

At each forecast origin on day D , the model only uses information available under the competition constraints—covariates observed up to midday on day D and outcome observations available only through day $D - 3$ —due to reporting delays. Therefore, the latent autoregressive state is conditioned on information up to day $D - 3$ and propagated forward when generating forecasts for days $D + 1$ through $D + 10$.

Mathematical Framework

Likelihood: We model the log1p-transformed observed avoidable deaths y_t on operational day t to be normally distributed around a time-varying total structural mean μ_t ,

$$\log(y_t + 1) \sim \text{Normal}(\mu_t, \sigma_{\text{obs}}).$$

The structural mean is an additive decomposition of seasonal, covariate, autoregressive memory effects, and day-of-week,

$$\mu_t = \mu_{\text{seasonal},t} + \mu_{\text{cov},t} + \mu_{\text{AR},t} + \phi_d(t),$$

where,

$$\begin{cases} \mu_{\text{seasonal},t} = \mu_0 + A \cdot \cos\left(\frac{2\pi \cdot t}{365.25} - \psi\right), \\ \mu_{\text{cov},t} = \sum_{l=0}^{-13} \sum_{c=1}^{59} \beta_{l,c} \cdot X_{t,l,c}, \\ \mu_{\text{AR},t} = \rho \cdot \mu_{\text{AR},t-1} + \epsilon_{\text{AR},t}. \end{cases}$$

Here, $\mu_{\text{cov},t}$ aggregates the regularized covariate effects over a lookback window of lags $l = -1, \dots, -13$ days across all 59 unique operational predictors.

Initial Condition: The initial state of the autoregressive memory component uses the empirical baseline residual on the first training day,

$$\mu_{\text{AR},0} \sim \text{Normal}\left(\theta_{\text{AR}}, \sqrt{\frac{\sigma_{\text{AR}}^2}{1 - \rho^2}}\right),$$

where,

$$\theta_{\text{AR}} = y_0 - (\mu_{\text{seasonal},0} + \mu_{\text{cov},0} + \phi_d(0)).$$

Priors and Parameter Constants

We assigned weakly informative priors to all model components to regularize estimation while preserving flexibility. The seasonal baseline parameters were centered on empirical summaries of the transformed outcome, where,

$$\begin{cases} \mu_0 \sim \text{Normal}(\hat{\mu}_0, \sigma = 1/3) & (\text{where } \hat{\mu}_0 \text{ is the log1p median of the observations}), \\ A \sim \text{LogNormal}(\hat{\mu}_A, \sigma = 1/3) & (\hat{\mu}_A \text{ is the estimated seasonal amplitude}), \\ \psi = 5.2 & (\text{fixed phase angle centering the peak around mid-January}). \end{cases}$$

Distributed-lag covariate effects were regularized using Laplace priors with lag-dependent scaling to control high-dimensional feature contributions, where,

$$\begin{cases} \beta_l^{\text{raw}} \sim \text{Laplace}(0, b_{\text{base}} = 1.0) & \forall l \in \mathcal{L}, c \in \mathcal{C}, \\ h(l) = e^{-\lambda|l|}, \\ b_l = h(l)^{\text{rev}} \cdot \left(\frac{0.015}{3\sqrt{2}} \right), \\ \beta_l = \beta_l^{\text{raw}} \cdot b_l. \end{cases}$$

where $\mathcal{L}$ and $\mathcal{C}$ represent the coordinate spaces for lags and covariate features, respectively, $\lambda = 0.1$ governs the exponential rate of decay across the lag space, and $h(l)^{\text{rev}}$ represents the reversed sequence of the exponential lag weights $h(l)$.

We used lag-dependent regularization to account for non-constant effects of covariates across the lag dimension. Specifically, our current approach applies stronger shrinkage to recent lags, prioritizing the signal from more distant lags, which remain available throughout the entire forecast horizon. Future iterations will evaluate alternative regularization structures on the assessment dataset to optimize the balance between short- and long-term predictive accuracy.

Day-of-week effects were modeled as zero-centered Gaussian deviations, where,

$$\phi_{d(t)} \sim \text{Normal}(0, \sigma = 0.05/3) \quad \forall d(t) \in \{\text{Mon}, \dots, \text{Sun}\}.$$

The latent memory system relies on a fixed daily persistence parameter,

$$\rho = 0.5.$$

Short-term shocks to the system are modeled as normally distributed innovations,

$$\epsilon_{AR,t} \sim \text{Normal}(0, \sigma_{AR}).$$

The scale of these random innovations is governed by a tightly constrained half-normal prior:

$$\sigma_{AR} \sim \text{HalfNormal}(\sigma = 0.01/3).$$

The observation noise scale represents the unexpected daily variance in observed admissions and is governed by a half-normal prior,

$$\sigma_{\text{obs}} \sim \text{HalfNormal}(\sigma = 0.05).$$

Data Preprocessing and Feature Engineering

The original dataset consisted of 16,728,656 rows and 6 columns, structured in a long format, containing covariate data at variable reporting frequencies, for several hospitals. Each row recorded an individual observation defined by the variables `dt`, `metric_name`, `coverage`, `value`, `coverage_label`, and `variable_type`.

All data was first assigned to its proper date, with any covariates recorded after noon on a given day being assigned to the subsequent day. This was done to respect the challenge guidelines, which state that prediction data for day D is only available up until midday. Next, we defined a heuristic to truncate the feature space by auditing each covariate to ensure it had at least 80% completeness temporally (at least one hospital has a complete entry at a given time), which reduced the initial feature space from 220 to 138 covariates. We then corrected for the fact that each covariate had varying reporting intervals, ranging from 15 minutes to one day. To resolve this, any data collected in shorter intervals was averaged to a single day based on its assigned operational date.

The long-format dataframe was then pivoted to a wide format. Each existing covariate was combined with its associated coverage label, creating distinct columns for each unique covariate belonging to each single hospital. This brought the number of covariates from 138 to 559. This wide data frame contained 930 observations, representing one for each day from March 16, 2023, to September 30, 2025.

Many predictors still had missing observations after the pivot, we set a 40% temporal completeness threshold to keep a covariate, which brought the number of covariates from 559 to 331. To further isolate the core drivers of hospital system strain and reduce the remaining dimensionality, we implemented a targeted regex whitelist filter. Columns were filtered based on specific operational keywords related to system stress, capacity bottlenecks, and flow delays, such as "OPEL", "occupancy", "Queue", "12 hours", and "60mins". We also explicitly whitelisted external features like localized admission numbers and regional emergency incident responses.

This text-matching filter retained 58 covariates derived from the NHS dataset. We then incorporated a regional influenza ICU admission indicator from the UKHSA South West surveillance system, yielding a final set of 59 predictors. The resulting analytic dataset comprised 930 daily observations and 61 columns, corresponding to the date, outcome variable, and 59 covariates [UK Health Security Agency, 2026].

Model Evaluation and Backtesting Performance

Model performance was evaluated using rolling-origin walk-forward backtesting to mirror the competition’s design. Over an evaluation period spanning August 16, 2025, through September 30, 2025, we executed 36 distinct out-of-sample evaluation windows (see Figures 1 and 2). For each forecast origin (day D), predictions for 10-days were generated. Out-of-sample Mean Squared Error (MSE) was calculated separately across two operational horizons: a short-term horizon (days $D + 1$ to $D + 5$) and a medium-term horizon (days $D + 6$ to $D + 10$).

The model achieved a mean MSE over the evaluation period of 0.0356 for days $D + 1$ to $D + 5$ and a mean MSE of 0.0482 for days $D + 6$ until $D + 10$. To quantify predictive skill, performance was benchmarked against a null model baseline constructed using a generalized additive model (GAM). We evaluated the performance ratio ($\text{MSE}_{\text{Model}}/\text{MSE}_{\text{Null}}$) across both horizons. The model substantially outperformed the baseline in both periods, achieving a relative MSE ratio of 0.5595 over the short-term horizon and 0.6392 over the medium-term horizon.

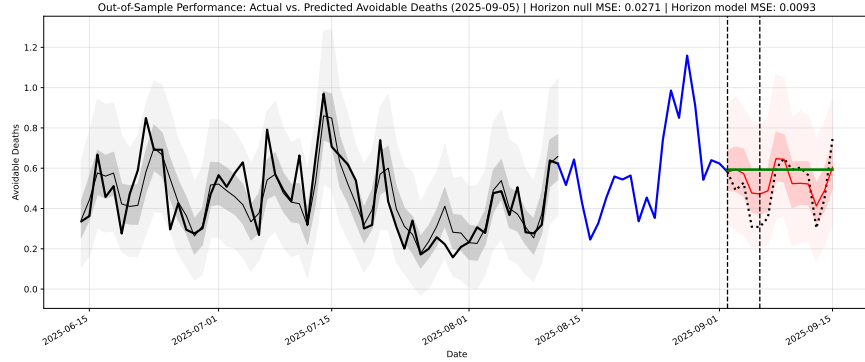


Figure 1: September 5, 2025 (day D). Our model’s median forecast is shown in red, along with its 50% and 95% forecast intervals. The right vertical line represents the forecast start date (D) while the left vertical line represent the data end date ($D - 3$). The green line represents the null model’s forecast.

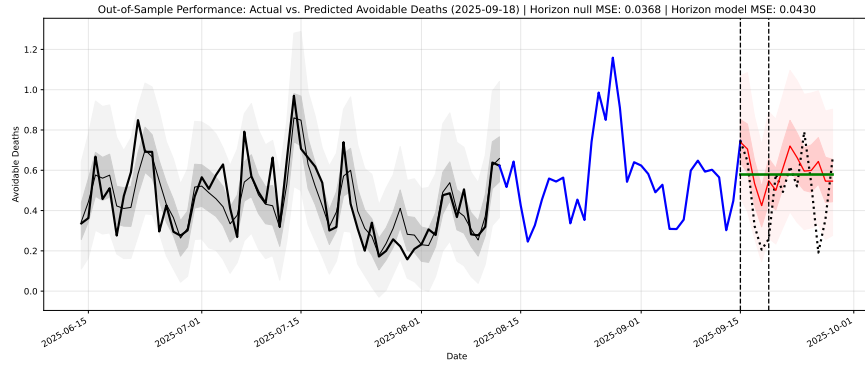


Figure 2: Out-of-Sample Performance starting September 18, 2025.

References

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